VLAN Trunking and Access Ports

Cisco Packet Tracer Lab Report

1. Theory: Trunking Protocols

- Access port: carries traffic for one VLAN only. Frames are untagged and the end device (PC) is unaware of VLANs.
- Trunk port: carries traffic for many VLANs between switches by adding a VLAN tag to each frame.
- IEEE 802.1Q: open standard, inserts a 4-byte tag into the Ethernet frame. Untagged traffic belongs to the native VLAN (default VLAN 1).
- ISL (Inter-Switch Link): Cisco proprietary, encapsulates the whole frame with a 26-byte header and 4-byte trailer. It is deprecated and not supported on newer switches.
- Both ends of a trunk must use the same encapsulation, or the link will not pass VLAN traffic.

Note for Packet Tracer: the 2960 switch supports only 802.1Q and rejects the 'switchport trunk encapsulation' command. Use the multilayer Cisco 3560-24PS for this lab so that both dot1q and ISL can be configured.

2. Topology and Addressing

Two switches (Switch1, Switch2) are joined with a crossover cable on FastEthernet0/1. Three PCs connect to Switch1. Two extra PCs on Switch2 are used to test traffic across the trunk.

Device	Switch Port	VLAN	IP Address	Mask
PC1	Switch1 Fa0/2	10	192.168.10.10	255.255.255.0
PC2	Switch1 Fa0/3	20	192.168.20.10	255.255.255.0
PC3	Switch1 Fa0/4	30	192.168.30.10	255.255.255.0
PC4	Switch2 Fa0/2	10	192.168.10.20	255.255.255.0
PC5	Switch2 Fa0/5	20	192.168.20.20	255.255.255.0

3. Task 1: Configure the Trunk (802.1Q)

Run on both switches (change the hostname on Switch2):

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname Switch1
Switch1(config)# interface fastEthernet 0/1
Switch1(config-if)# switchport trunk encapsulation dot1q
Switch1(config-if)# switchport mode trunk
Switch1(config-if)# end
```

Verify with: show interfaces trunk

4. Task 2: Create VLANs and Assign Access Ports

Create the VLANs on both switches:

```
Switch1(config)# vlan 10
Switch1(config-vlan)# name VLAN10
Switch1(config-vlan)# vlan 20
Switch1(config-vlan)# name VLAN20
Switch1(config-vlan)# vlan 30
Switch1(config-vlan)# name VLAN30
Switch1(config-vlan)# exit
```

Assign the access ports on Switch1:

```
Switch1(config)# interface fastEthernet 0/2
Switch1(config-if)# switchport mode access
Switch1(config-if)# switchport access vlan 10
Switch1(config-if)# interface fastEthernet 0/3
Switch1(config-if)# switchport mode access
Switch1(config-if)# switchport access vlan 20
Switch1(config-if)# interface fastEthernet 0/4
Switch1(config-if)# switchport mode access
Switch1(config-if)# switchport access vlan 30
Switch1(config-if)# end
```

On Switch2, put Fa0/2 in VLAN 10 and Fa0/5 in VLAN 20 as access ports in the same way.

Verify with: show vlan brief

5. Task 3: Test Connectivity

Test (ping from, to)	Expected	Reason
PC1 to PC2 (VLAN 10 to 20)	Fails	Different VLANs, no router
PC1 to PC3 (VLAN 10 to 30)	Fails	Different VLANs, no router
PC2 to PC3 (VLAN 20 to 30)	Fails	Different VLANs, no router
PC1 to PC4 (VLAN 10, across trunk)	Succeeds	Same VLAN, trunk carries VLAN 10
PC2 to PC5 (VLAN 20, across trunk)	Succeeds	Same VLAN, trunk carries VLAN 20

Because all three PCs on Switch1 are in different VLANs, same-VLAN communication is shown with the extra PCs on Switch2 across the trunk. Allow the first ping a few seconds (ARP) before judging results. Communication between different VLANs would need a router (router-on-a-stick) or a Layer 3 switch.

6. Task 4: Change the Trunk to ISL

On both switches, set the encapsulation before the mode:

```
Switch1(config)# interface fastEthernet 0/1
Switch1(config-if)# switchport trunk encapsulation isl
Switch1(config-if)# switchport mode trunk
Switch1(config-if)# end
Switch1# show interfaces trunk
```

Results to record:

- Both ends set to ISL: the trunk comes up (encapsulation shows isl) and PC1 to PC4 still succeeds. VLAN traffic passes.
- Only one end set to ISL (the other still dot1q): the encapsulations do not match, the trunk fails to pass VLAN traffic, and PC1 to PC4 fails. Fix by matching both ends.
- If the switch rejects the command, it is a 2960, which supports only 802.1Q. Replace it with a 3560.

7. Task 5: Fa0/5 Accidentally Set as a Trunk

The fault on Switch2:

```
Switch2(config)# interface fastEthernet 0/5
Switch2(config-if)# switchport trunk encapsulation dot1q
Switch2(config-if)# switchport mode trunk
```

Impact on PC5 (192.168.20.20), which was in VLAN 20:

- A trunk port expects tagged frames and carries all VLANs. A PC sends untagged frames, so the switch places them in the native VLAN (VLAN 1), not VLAN 20.
- PC5 can no longer reach PC2 or any other VLAN 20 device. Pings fail even though the link light is green.
- PC5 would also receive traffic for VLAN 1 and, with an unrestricted trunk, is exposed to frames from all VLANs. This is a security risk (VLAN hopping).
- If PC5 used DHCP, it would not get an address from the VLAN 20 pool.

Fix:

```
Switch2(config)# interface fastEthernet 0/5
Switch2(config-if)# switchport mode access
Switch2(config-if)# switchport access vlan 20
```

After the fix, PC5 to PC2 succeeds again.

8. Conclusion

Trunk ports carry multiple VLANs between switches using 802.1Q (or ISL), while access ports serve a single VLAN for end devices. Both trunk ends must use the same encapsulation, and a PC port wrongly set as a trunk loses its VLAN membership and connectivity.